S9 Joint comparison with algebraic methods

Figures S1 and S2 place the recorded DFO and algebraic results on the same completion-time axis. DFO times are reconstructed from the stored search paths and separately measured configuration costs; algebraic times are measured and include preparation. Each DFO condition has ten seeds, whereas each algebraic method has one run per problem. The original target rules are retained: absolute agreement with the reference for DFO, and a feasible objective at or below the reference for algebraic methods, using the stated tolerance. This descriptive comparison does not estimate an algebraic success probability.

Table S8: NLP workload in the recorded process comparisons. DFO entries are median counts of corresponding fixed-configuration NLP evaluations over ten seeds; GDPopt entries count confirmed SCIP NLP interface calls in one run. Logical and preprocessing rejections are excluded; rejected NLP returns are retained. These counts are not internal solver iterations or equally costly subproblems.

Case	B	D-SDA	RBFOpt	BAM	FMQA-SA	Uniform	GLOA	LOA	RIC
CSTR5	8	6	6	6	5.5	4.5	2	15	15
CSTR10	15	10	12	8.5	9	7.5	2	55	55
CSTR15	22	17	16.5	11	15	11	2	120	120
CSTR20	29	21	20.5	17	17.5	15.5	2	210	210
CSTR25	36	25	25.5	18.5	23	21	2	325†	325
CSTR30	43	26.5	31.5	21.5	30.5	21.5	2	362	370†
Column	13	11	12	7.5	7	7	1	1	1
Small Batch	10	4.5	6	5	5	4	12	12	12
Dynamic	12	10	10	8	6.5	6	1	1	1
Dynamic	15	13	13	9	8	6.5	1	1	1
Dynamic	20	17.5	16	11	11	9.5	1	1	1
Dynamic	30	22	21	17	14.5	15	1	1	1
Dynamic	37	22	25	19	19	19	1	1	1
Dynamic	50	22	31	26	24.5	26	1	1	1

The same single Dynamic GDPopt execution is repeated across budget rows for reference. †CSTR25 LOA and CSTR30 RIC each have one additional NLP interface attempt without a confirmed solver start; these attempts are excluded. Internal NLP counts for direct Big-M and Hull are unavailable and are not treated as zero.

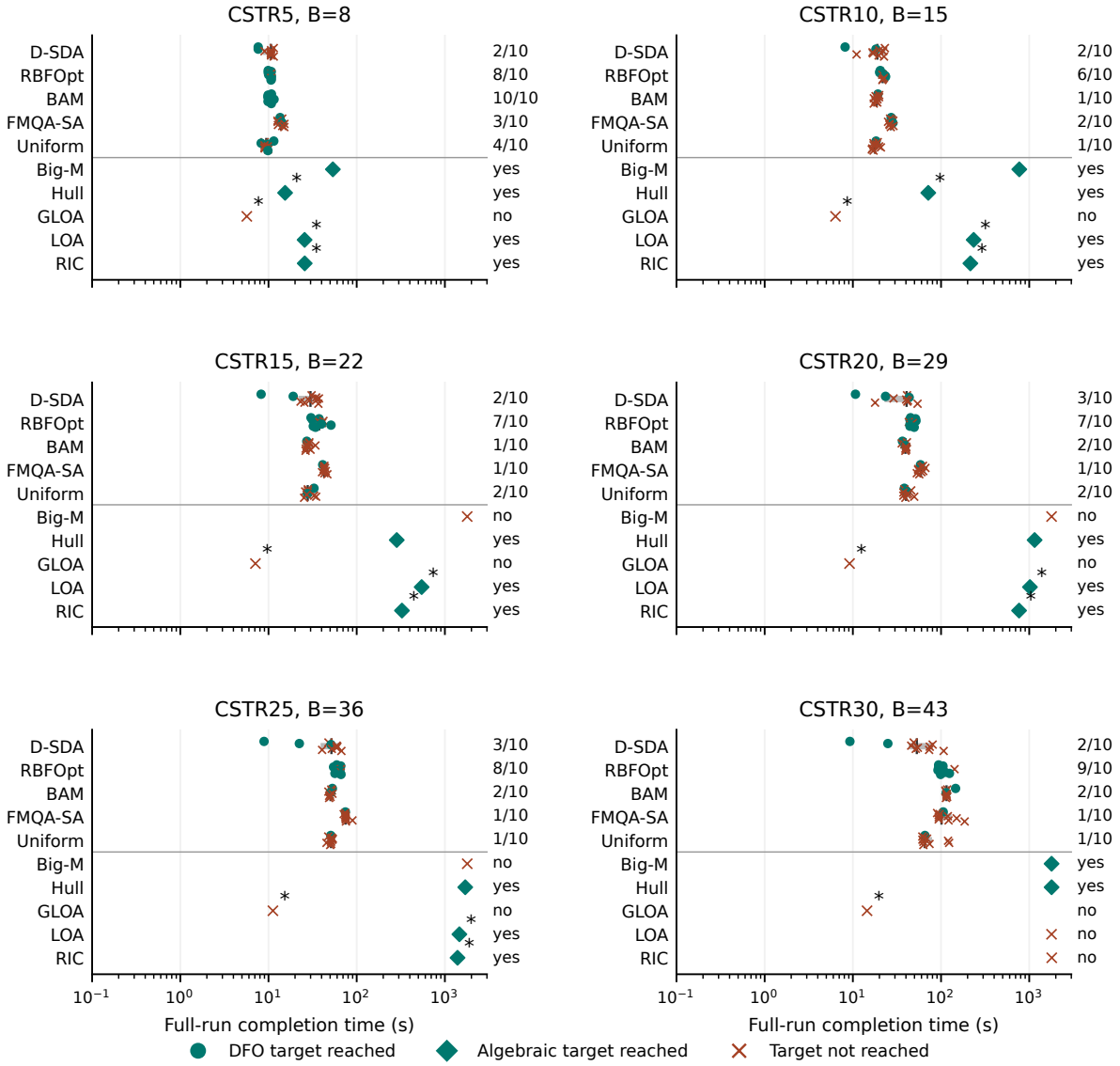


Figure S1: Joint completion-time comparison for the six CSTR cases. Each DFO dot represents one seed; gray segments show Q25–Q75 and vertical ticks show medians across all ten runs. Diamonds represent successful single algebraic runs; crosses indicate target misses in either group. Right-hand labels give DFO hits out of ten or the algebraic yes/no outcome. Asterisks retain the bound-inconsistency flags discussed in Section S7. The horizontal axis is logarithmic and shows full-run completion, not first-target time.

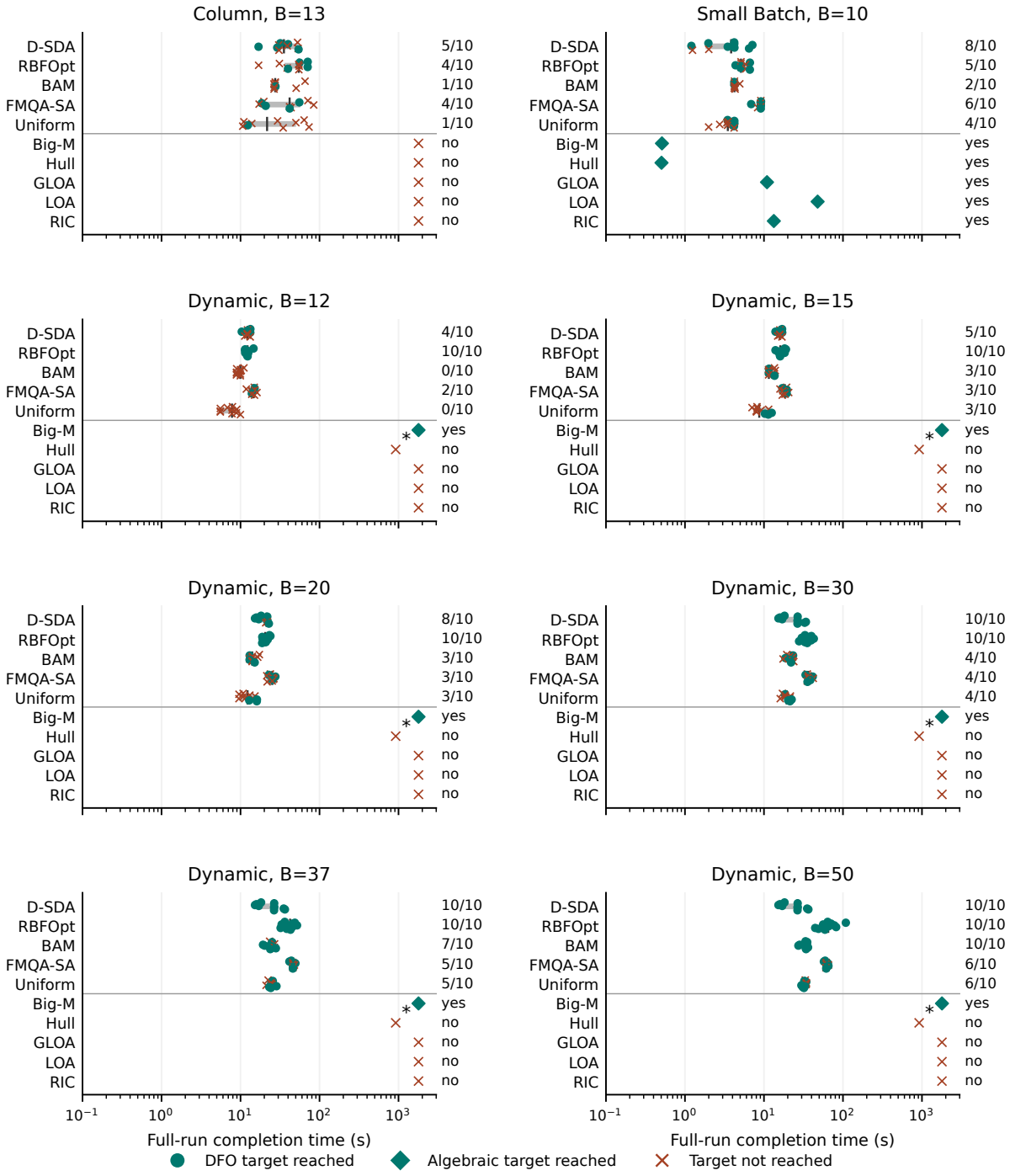


Figure S2: Joint completion-time comparison for Column, Small Batch, and the six Dynamic budgets. Each DFO dot represents one seed; gray segments show Q25–Q75 and vertical ticks show medians across all ten runs. Diamonds represent successful single algebraic runs; crosses indicate target misses in either group. Right-hand labels give DFO hits out of ten or the algebraic yes/no outcome. Asterisks retain the bound-inconsistency flags discussed in Section S7. The horizontal axis is logarithmic and shows full-run completion, not first-target time. The same single algebraic Dynamic run appears in each budget panel; these are not additional runs.